SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning,Data intergration,Data transformation and data Reduction ,for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding *(2)*	Method Selection *(2)*	Solution Process *(2.5)*	Accuracy of Calculation *(2)*	Interpretation of Results *(1.5)*	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I M. Pranay (192525382)_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
M. Pranay

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B
Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	**Claims per 100 Policyholders (yi)**
3.2	15
4.0	18
5.5	25
4.8	20

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2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

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3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

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4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

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5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

1) Suppose data for analysis include the attribute age. The age values for the data tuple or in increasing order.

13, 15, 16, 16, 19, 20, 20, 21
22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45,
46, 52, 70.

Sol:-

(i) what is the mean of Data & what is the median
(ii) what is the mode of data.
unimodel Bimodal, trimodel.

(iii) what is the midrange of the data.
(iv) can you find first quartile (Q_1) and third quartile (Q_3) of the data.
(v) Give the 5 numbers summary of the data.
(vi) Show the bDx Plot of the data.

Sol:-

(i) $\text{Mean} = \frac{\text{sum of all the data}}{\text{no. of data}} = \frac{809}{27} = 29.96 = 30$.

$\text{Median} = \frac{27+1}{2} = 14^{\text{th}} \text{ index} = 25$.

iii) 25 occuring for four times and also 235 occuring for four times.
So, Bothe 25 & 35 are the repeated numbers of the age

So, it is bimodel.

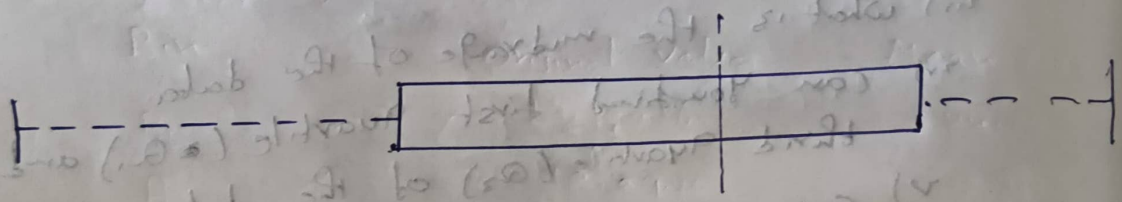
iii) Midrange = $\frac{\text{minvalue} + \text{Maxvalue}}{2}$ = $\frac{13+70}{2}$ = 41.5

iv) 13, 15, 16, 19, 20, 20, 21, 22, 23, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.

$Q_1 = \frac{1+13}{2}$ = 7 index = 20

$Q_3 = \frac{15+27}{2}$ = 21 index = 35

v) (MIN, Q_1, MEDIAN, Q_3, MAX)
= (13, 20, 25, 35, 70).



13 15 16 19 20 21 22 25 30 33 35 36 40 45 46 52 70

2) Give the following measurments for the variable age:- 18, 22, 25, 42, 28, 43, 33, 35, 56, 28.

Compute the mean absolute deviation (mad).

$$\text{Mean} = \frac{330}{10} = 33.$$

X (age)	$\lvert x - \bar{x} \rvert$ (Mean)	Calculate
18	18-33	15
22	22-33	11
25	25-33	8
42	42-33	9
28	28-33	5
43	43-33	10
33	33-33	0
35	35-33	2
56	56-33	23
28	28-33	5
		88

$$MAD = \frac{\sum |x - \bar{x}|}{N} = \frac{88}{10} = 8.8.$$

3) Suppose the data for ~~analysis~~ analysis include the attribute age the age value for the data tuples are in increasing order.

13, 15, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Using smoothing by bin means to smooth the above data using bindepth of 3. Illustrate your steps.

step 1:-

Soln: Mean = $\frac{809}{27} = 30$.

step 2:- No. of occurances = 27

BinDepth = 3

Count of bin = $\frac{\text{Occurance}}{\text{depth}} = \frac{27}{3} = 9$.

step 3:-

Bin	Original values	Mean (M)
B_1	13, 15, 16	44.66
B_2	19, 20, 20	18.33
B_3	20, 21, 22	21
B_4	22, 25, 25	24
B_5	25, 30, 30	26.66
B_6	33, 33, 35	33.66
B_7	35, 35, 35	35
B_8	36, 40, 45	40.33
B_9	46, 52, 70	56

step 4 :-

~~Replace c~~

Replace calculated mean for all three values in the bin.

Bin	Smoothed Values.
B_1	44.66, 44.66, 44.66
B_2	18.33, 18.33, 18.33
B_3	21, 21, 21
B_4	24, 24, 24
B_5	26.66, 26.66, 26.66
B_6	33.66, 33.66, 33.66
B_7	35, 35, 35
B_8	40.33, 40.33, 40.33
B_9	56, 56, 56.

26/10/20

4) Use two methods given to you to normalize following group of data:- 200, 300, 400, 600, 1000

(a) Mean max normalization by assigning min=0; max=1

(b) Z-score normalization

Soln: (a) $\bar{x} = \frac{x - \min(x)}{\max(x) - \min(x)}$

$\max(x) - \min(x)$

Original value (x)	Calculation	normalization ($\bar{x}$)
200	$= \frac{200-200}{800}$	0
300	$= \frac{300-200}{800}$	0.125
400	$= \frac{400-200}{800}$	0.25
600	$= \frac{600-200}{800}$	0.5
1000	$= \frac{1000-200}{800}$	1

(b) Z score normalization:-

step 1:- $Z = \frac{x - \mu}{\sigma}$

$\text{mean} = \frac{200+300+400+600+1000}{5}$

$= 500.$

step 2:- Calculation of standard deviation:-

x	$\lvert x-\mu \rvert$	$(x-\mu)^2$
200	300	90000
300	200	40000
400	100	10000
600	100	10000
1000	500	250000
		400,000

(σ^2) • Variance $= \frac{1}{N} * \Sigma (x-\mu)^2$

$= \frac{1}{5} (400000)$

$\sigma^2 = 80,000.$

$\sigma = \sqrt{80,000} = 282.84$

Step 3:- Calculating Z-score:-

x	$\frac{x-\mu}{\sigma}$
200	$\frac{-300}{282.84} = -1.06$
300	$\frac{-200}{282.84} = -0.707$
400	$\frac{-100}{282.84} = -0.353$
600	$\frac{100}{282.84} = 0.353$
1000	$\frac{500}{282.84} = 1.767$

Values of Z :- $-1.06, -0.707, -0.353, 0.353, 1.767$

5) Suppose a group of 12 sales price records has been sorted as follows :- 5, 10, 11, 13, 15, 35, 15, 55, 72, 92, 204, 215.

Partition them into three bins by each of the following methods.

1) Equal frequency Partition
2) Equal width Partition
3) Clustering.

Soln:- Total records = 12

Bins = 3

(i) Equal frequency Partition

$$\text{Frequency} = \frac{\text{record}}{\text{Bin}} = \frac{12}{3} = 4.$$

Bin	values
B_1	5, 10, 11, 13
B_2	15, 35, 15, 55
B_3	72, 92, 204, 215

(ii) Equal width Partition.

$$\text{width} = \frac{\text{Max} - \text{Min}}{\text{Bin}} = \frac{215 - 5}{3} = 70.$$

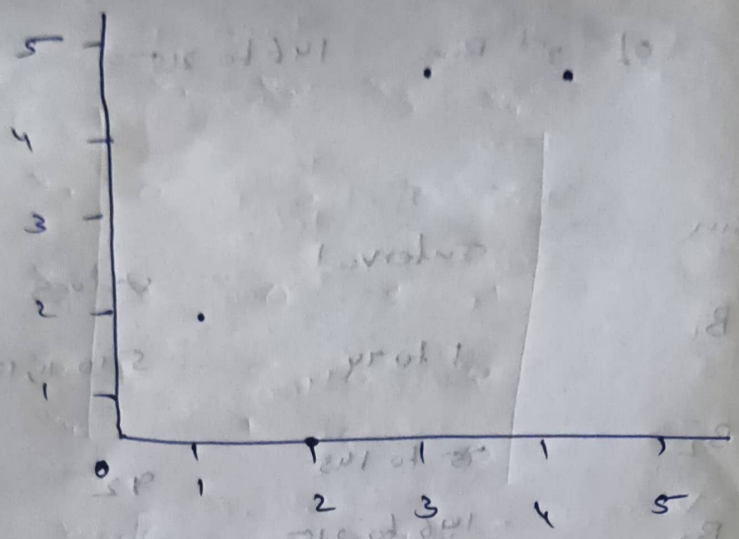
Interval of 1st Bin 1 to 74
Interval of 2nd Bin 75 to 145
Interval of 3rd Bin 146 to 215

Bin	Interval	values
B_1	1 to 74	5, 10, 11, 13, 15, 35, 55, 72
B_2	75 to 145	92
B_3	146 to 215	204, 215

iii) clustering:-

Cluster (or) Bin	values
Bin 1	5, 10, 11, 13, 15
Bin 2	35, 55, 72, 92
Bin 3	204, 215

6) Consider the following Points:-
(1,2), (3,5), (2,0) & (4,5) construct the data matrix & Similarity matrix. Manhattan, Euclidean Distance, Supremum.



Data matrix:-

	Attribute 1	Attribute 2
x_1	1	2
x_2	3	5
x_3	2	0
x_4	4	5

ii) Dissimilarity matrix manhattan:-

	x_1	x_2	x_3	x_4
x_1	0	5	3	6
x_2	5	0	6	1
x_3	3	6	0	7
x_4	6	1	7	0

Distance blw x_2 and x_1 :- (3,5) and (1,2)

$= |(x_1-x_2)| + |(y_1-y_2)|$

$= 2+3 = 5$

Distance blw x_3 (2,0) and x_1 (1,2)

$= 1+2 = 3$

Distance blw x_4 (4,5) and x_1 (1,2)

$= 3+3 = 6$

Distance blw x_1 (1,2) and x_2 (3,5)

$= 2+3 = 5$

Distance blw x_3 (2,0) and x_2 (3,5)

$= 1+5 = 6$

Distance blw x_4 (4,5) and x_2 (3,5)

$= 1+0 = 1$

Distance blw x_4 (4,5) and x_3 (2,0)

$= 2+5 = 7$

2.iv) euclidean Distance:-

$$\sqrt{(x_1-x_2)^2 + (y_1-y_2)^2}$$

	x_1	x_2	x_3	x_4
x_1	0	3.605	2.23	4.24
x_2	3.605	0	5.09	1
x_3	2.23	5.09	0	5.38
x_4	4.24	1	5.38	0

$x_2, x_1 = \sqrt{(4+9)} = \sqrt{13} = 3.605$

$x_3, x_1 = \sqrt{1+4} = \sqrt{5} = 2.23$

$x_4, x_1 = \sqrt{9+9} = \sqrt{18} = 4.24$

$x_3, x_2 = \sqrt{1+25} = \sqrt{26} = 5.09$

$x_4, x_2 = \sqrt{1+0} = 1$

$x_4, x_3 = \sqrt{4+25} = \sqrt{29} = 5.38$

iv) Dissimilarity matrix using Supermarket

	x_1	x_2	x_3	x_4
x_1	0	3	2	3
x_2	3	0	5	1
x_3	2	5	0	5
x_4	3	1	5	0

$d = \max[(x_1 - x_2), (y_1 - y_2)]$

$x_2, x_1 = 3$.
$x_3, x_1 = 2$
$x_4, x_1 = 3$

$x_3, x_2 = 5$
$x_4, x_2 = 1$
$x_4, x_3 = 5$

7) Finding the similarity measure b/w two documents using cosine similarity.

Document	team	coach	hockey	baseball	soccer	Penalty
1	5	0	3	0	2	0
2	3	0	2	0	1	1
3	0	7	0	2	1	0
4	0	1	0	0	1	2

Score	win	loss	season
0	2	0	0
0	1	0	1
0	3	0	0
2	0	3	0

$$\cos(d_1, d_2) = \frac{d_1 \cdot d_2}{|d_1| \cdot |d_2|}$$

$d_1 = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$

$d_2 = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$

$d_1 \cdot d_2 = (15, 0, 6, 0, 2, 0, 0, 2, 0, 0)$

$= (15, 6, 2, 2)$

$= 25.$

$|d_1| = (5, 3, 2, 2)$

$= (25, 9, 4, 4)$

$= \sqrt{42}$

$= 6.48$

$|d_2| = (3, 2, 1, 1, 1, 1)$

$= 9(9, 4, 1, 1, 1, 1)$

$= \sqrt{17}$

$= 4.12$

$$\cos(d_1, d_2) = \frac{(d_1 \cdot d_2)}{|d_1| \, |d_2|} = \frac{25}{(6.48)(4.12)} = 0.936.$$

28/10/26